

Unit V: Fourier Transform

Assignment: Properties of Fourier Transforms:

- 1) Find the Fourier Transform for the function $f(x) = \begin{cases} 6x, & |x| \leq a \\ 0, & |x| > a \end{cases}$ and hence find $F[e^{7ix}f(x)]$
- 2) Find the Inverse Fourier Transform for the function $F(s) = \frac{5}{(2+is)} + \frac{4is+10}{(2-is)^2}$